

Wi-Fi Signal Optimization

PSO Router Placement - WillIndoorLoc

Upload Dataset

TrainingData.csv

TrainingData.csv
10.7MB

ValidationData.csv (optional)

ValidationData.csv
2.3MB

Signal Settings

Floor to analyze

All Floors

Weak signal threshold (dBm)

-65

PSO Parameters

Routers to place

3

Number of particles

10

Wi-Fi Signal Coverage Optimization

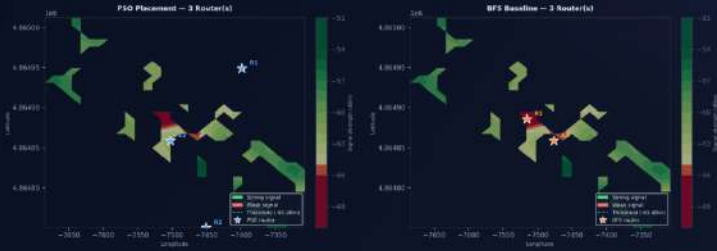
PSO-based router placement · All Floors · 19,937 training samples · Weak zone threshold: -65 dBm

CURRENT SIGNAL COVERAGE

WEAK ZONE AREA ⓘ	AVERAGE SIGNAL	BEST SIGNAL	WORST SIGNAL
7.5%	-59.9 dBm	-51.9 dBm	-70.5 dBm

PSO OPTIMIZATION RESULTS

WEAK ZONES BEFORE	WEAK ZONES AFTER PSO	PSO COVERAGE GAIN	BFS COVERAGE GAIN	PSO IMPROVEMENT OV...
7.5%	0.0% ↑ -7.5%	100.0%	100.0%	+0.0%



PSO Parameters

Routers to place

3

Number of particles

10

Number of iterations

8

Inertia weight (w)

0.70

Cognitive coeff (c1)

1.50

Social coeff (c2)

1.50

RUN PSO

Live Simulation

Enable live floor prediction

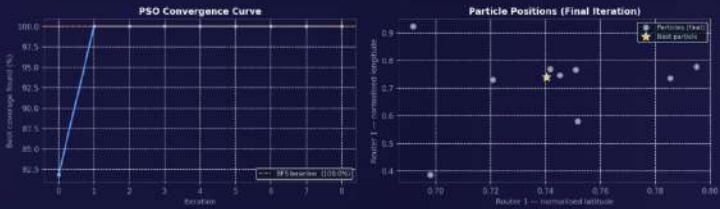
Refresh (s)

3

Signal noise (dBm)

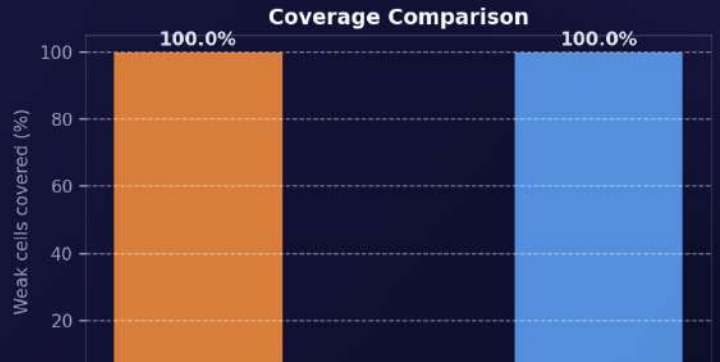
5

PSO CONVERGENCE ANALYSIS



Left: blue curve shows how PSO improves each iteration. Flat sections = swarm stagnation (no better position found). Right: where each particle ended up in the search space at the final iteration.

PSO VS BFS — COVERAGE COMPARISON



PSO Parameters

Routers to place

3

Number of particles

10

Number of iterations

8

Inertia weight (w)

0.70

Cognitive coeff (c1)

1.50

Social coeff (c2)

1.50

RUN PSO

Live Simulation

☒ Enable live floor prediction

Refresh (s)

3

Signal noise (dBm)

5



Router placement coordinates

LIVE FLOOR PREDICTION — SIGNAL MONITORING



PSO Parameters

Routers to place

Number of particles

Number of iterations

Inertia weight (w)

Cognitive coeff (c1)

Social coeff (c2)

▶ RUN PSO

Live Simulation

☐ Enable live floor prediction

Refresh (s)

Signal noise (dBm)

0

BFS (greedy)

PSO (p=10, i=8)

> Router placement coordinates

LIVE FLOOR PREDICTION — SIGNAL MONITORING

Enable live floor prediction in the sidebar to start the simulation.

DATASET & SIGNAL SOURCE ANALYSIS

ROUTERS PER FLOOR

SIGNAL DISTRIBUTION

TOP 5 ACCESS POINTS